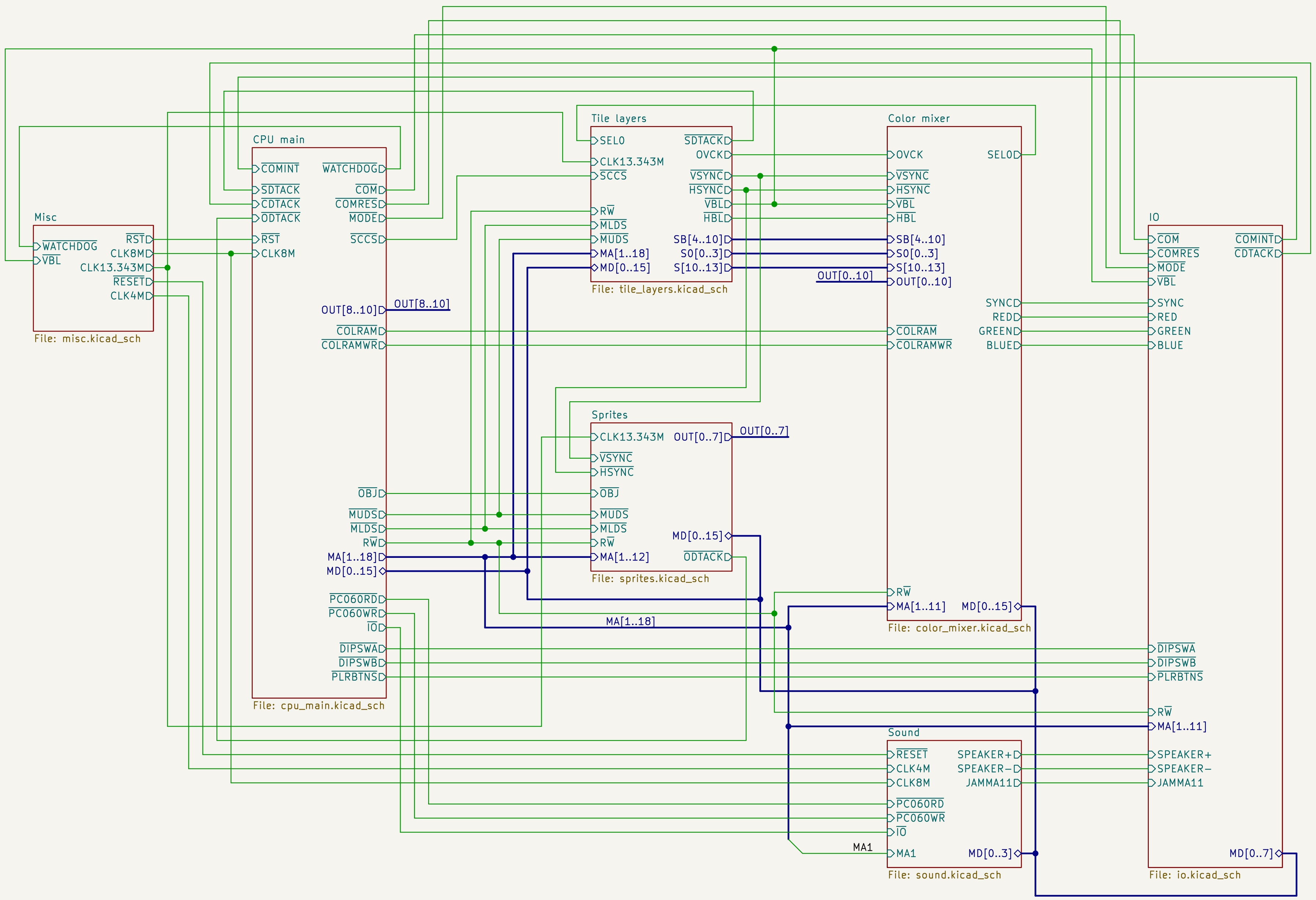

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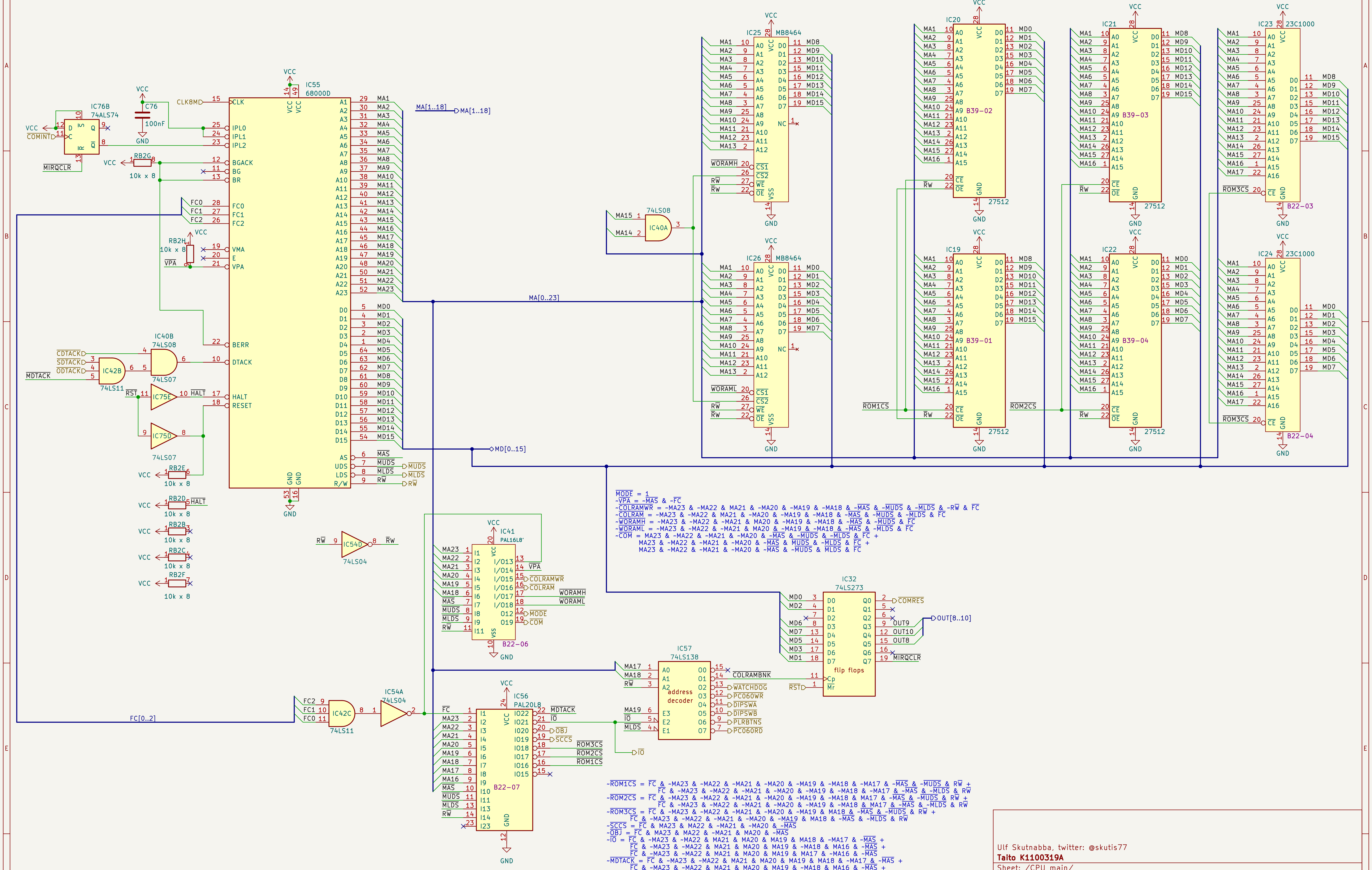
Taito K1100319A

Sheet: /

File: risle.kicad_sch

Title: Rainbow Islands Extra

Size: A3	Date: 2023-06-13	**Rev:**
KiCad E.D.A. 8.0.9		Id: 1/8




```
MODE = 1
~VPA = ~MAS & ~FC
~COLRAMWR = ~MA23 & ~MA22 & MA21 & ~MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS & ~MLDS & ~RW & FC
~COLRAM = ~MA23 & ~MA22 & MA21 & ~MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS & ~MLDS & FC
~WORAMH = ~MA23 & ~MA22 & ~MA21 & MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS & FC
~WORAML = ~MA23 & ~MA22 & ~MA21 & MA20 & ~MA19 & ~MA18 & ~MAS & ~MLDS & FC
~COM = MA23 & ~MA22 & ~MA21 & ~MA20 & ~MAS & ~MUDS & ~MLDS & FC +
       MA23 & ~MA22 & ~MA21 & ~MA20 & ~MAS & MUDS & ~MLDS & FC +
       MA23 & ~MA22 & ~MA21 & ~MA20 & ~MAS & ~MUDS & MLDS & FC
```

```
~ROM1CS = FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MA18 & ~MA17 & ~MAS & ~MUDS & RW +
          FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MA18 & ~MA17 & ~MAS & ~MLDS & RW
~ROM2CS = FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MA18 & MA17 & ~MAS & ~MUDS & RW +
          FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MA18 & MA17 & ~MAS & ~MLDS & RW
~ROM3CS = FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & MA18 & ~MAS & ~MUDS & RW +
       FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & MA18 & ~MAS & ~MLDS & RW
~SCCS = FC & MA23 & MA22 & ~MA21 & ~MA20 & ~MAS
~OBJ = FC & MA23 & MA22 & ~MA21 & MA20 & ~MAS
~IO = FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & MA18 & ~MA17 & ~MAS +
       FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & ~MA18 & MA16 & ~MAS +
       FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & MA17 & ~MA16 & ~MAS
~MDTACK = FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & MA18 & ~MA17 & ~MAS +
       FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & ~MA18 & MA16 & ~MAS +
       FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & MA17 & ~MA16 & ~MAS +
       FC & ~MA23 & ~MA22 & ~MA21 & MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS +
       FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MAS & ~MUDS & RW +
       FC & ~MA23 & ~MA22 & ~MA21 & MA20 & ~MA19 & ~MA18 & ~MAS & ~MLDS +
       FC & ~MA23 & ~MA22 & MA21 & ~MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS & ~MLDS +
       FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MAS & ~MLDS & RW
```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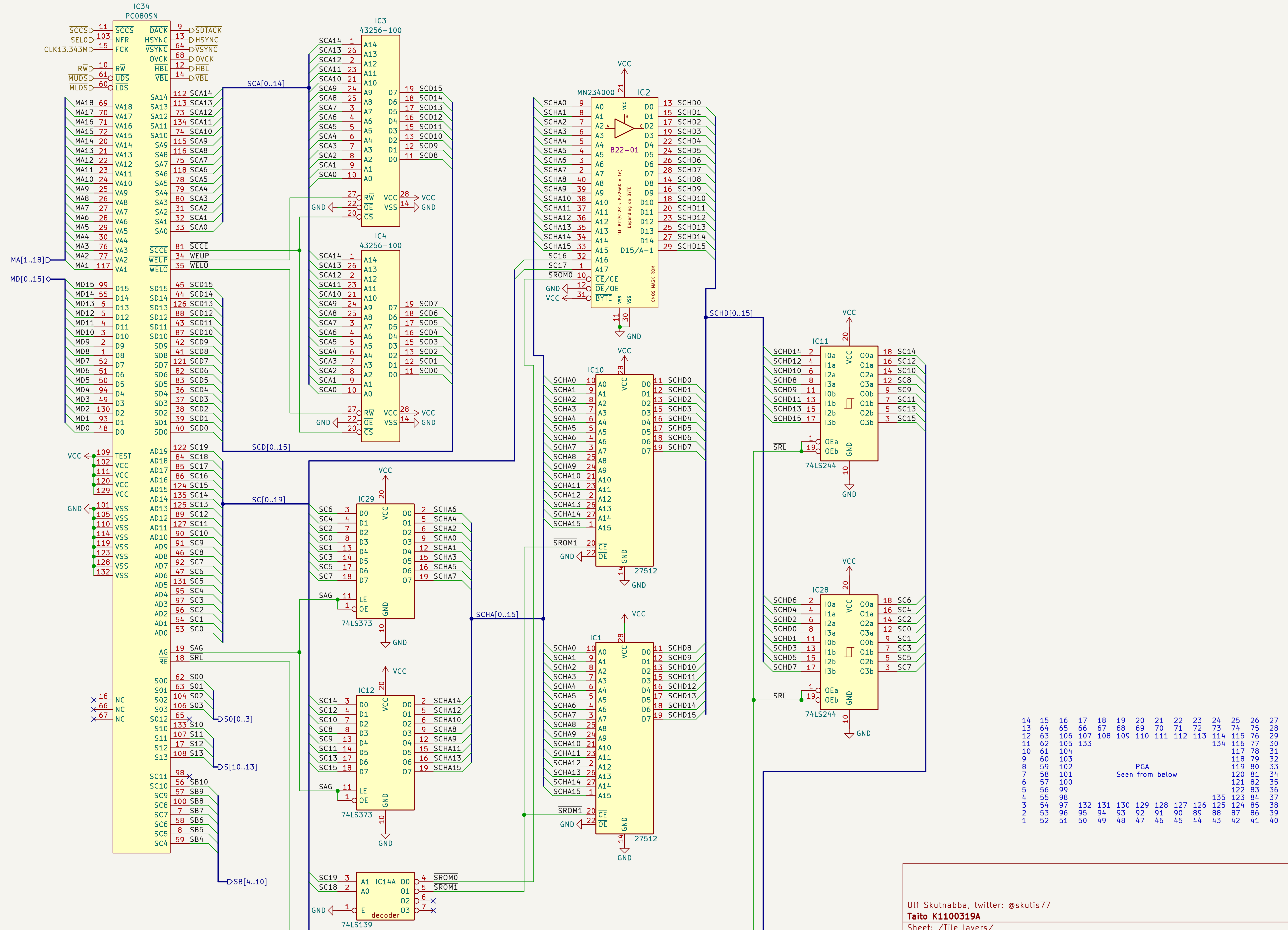
Sheet: /CPU main/

File: cpu_main.kicad_sch

Title: Rainbow Islands Extra

Size: A3 | Date: 2023-06-13 | Rev:

KiCad E.D.A. 8.0.9 | Id: 2/8



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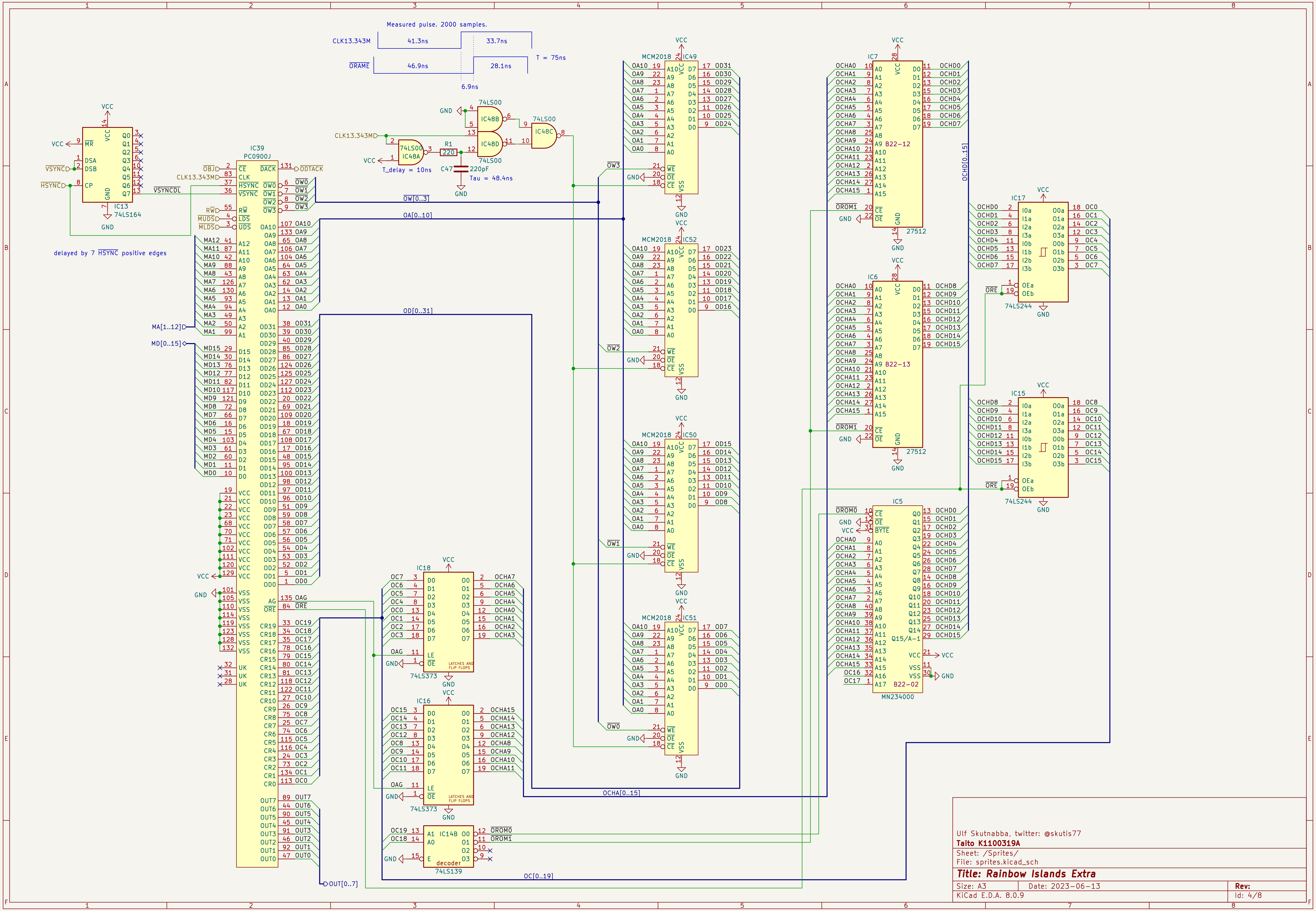
Sheet: /Tile layers/

File: tile_layers.kicad_sch

Title: Rainbow Islands Extra

Size: A3 | Date: 2023-06-13 | Rev:

KiCad E.D.A. 8.0.9 | Id: 3/8



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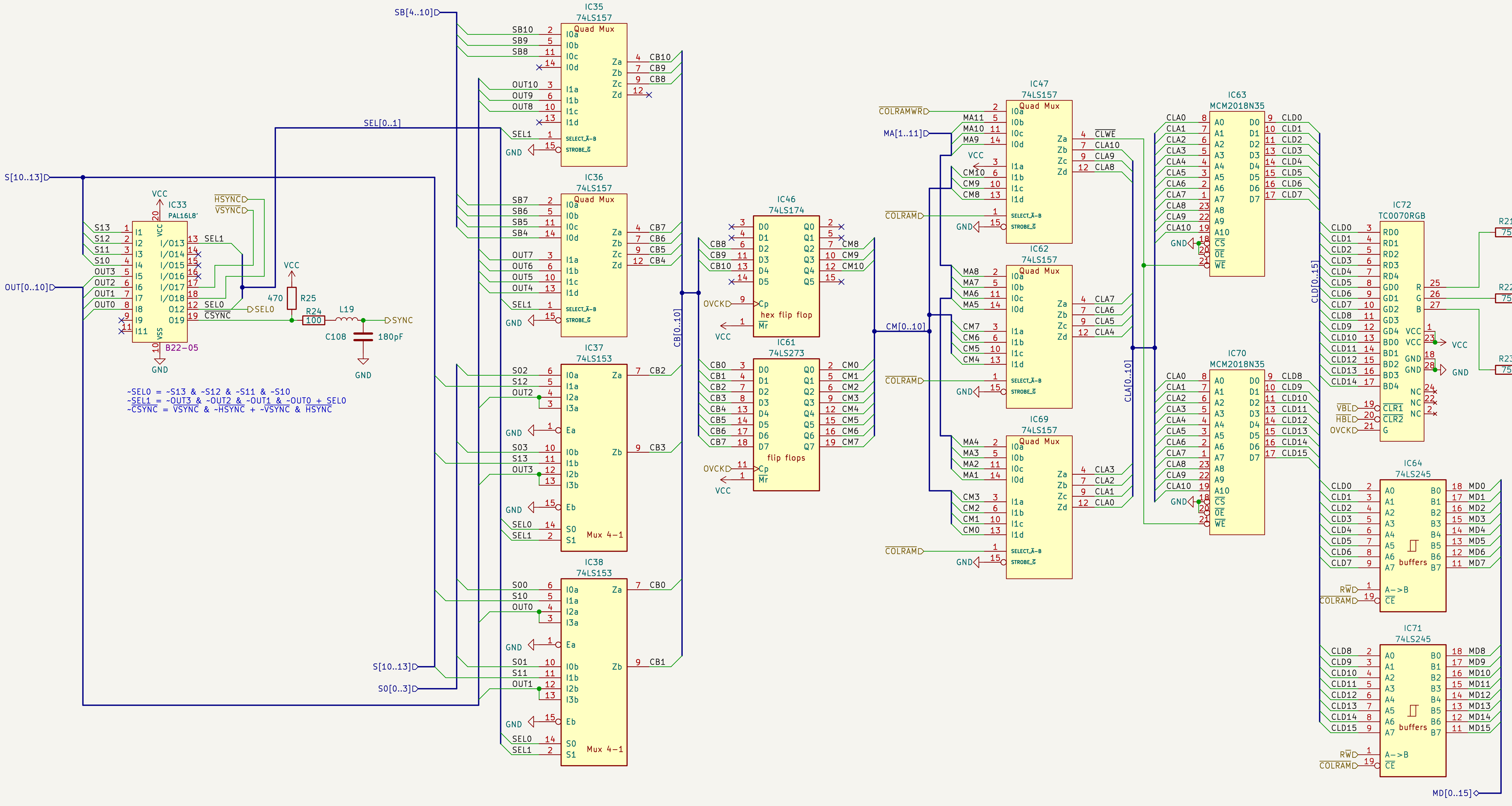
Sheet: /Sprites/

File: sprites.kicad_sch

Title: Rainbow Islands Extra

Size: A3 | Date: 2023-06-13 | Rev:

KiCad E.D.A. 8.0.9 | Id: 4/8



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Taito K1100319A

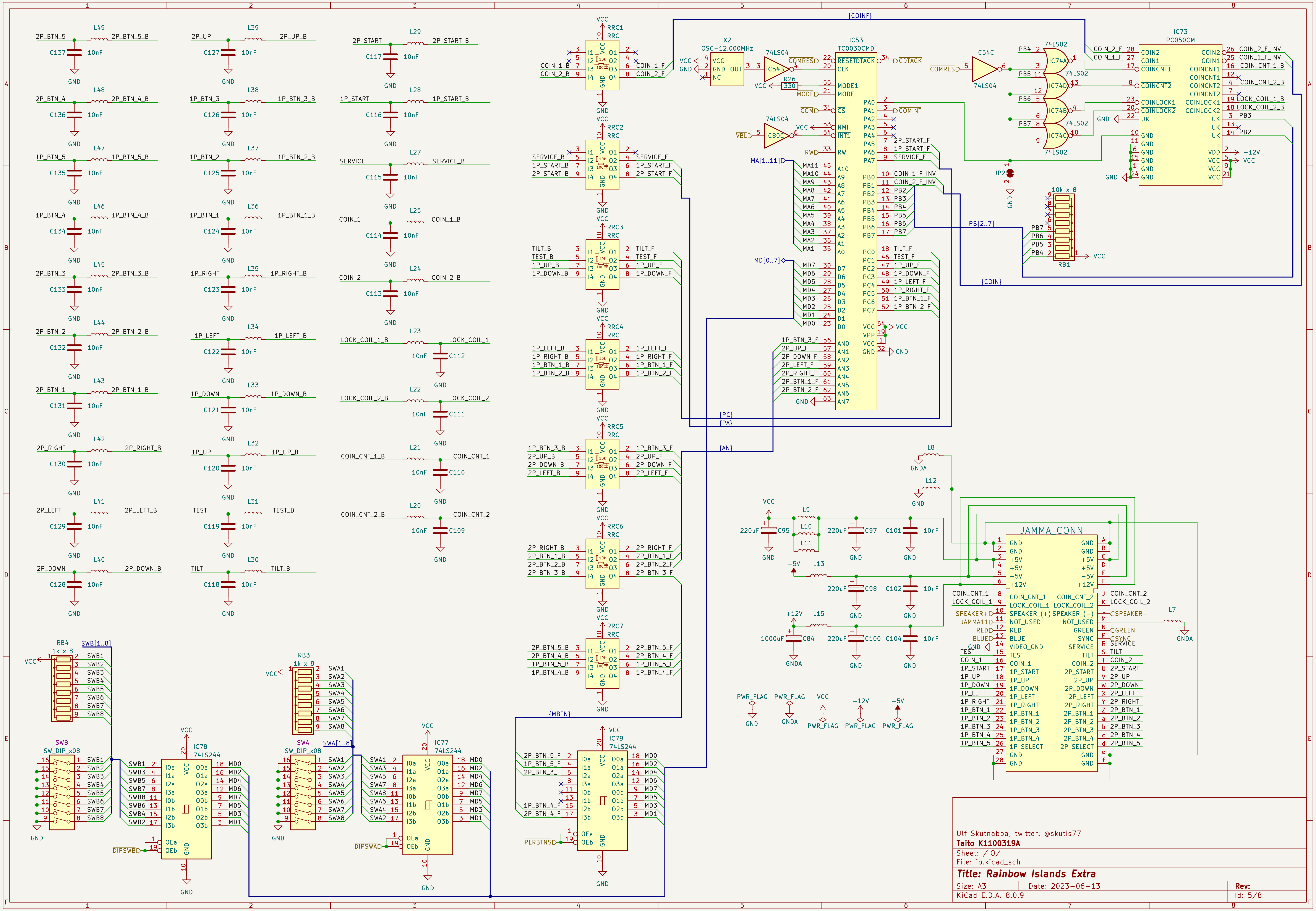
Sheet: /Color mixer/

File: color_mixer.kicad_sch

Title: Rainbow Islands Extra

Size: A3 Date: 2023-06-13 **Rev:**

KiCad E.D.A. 8.0.9 Id: 4/8



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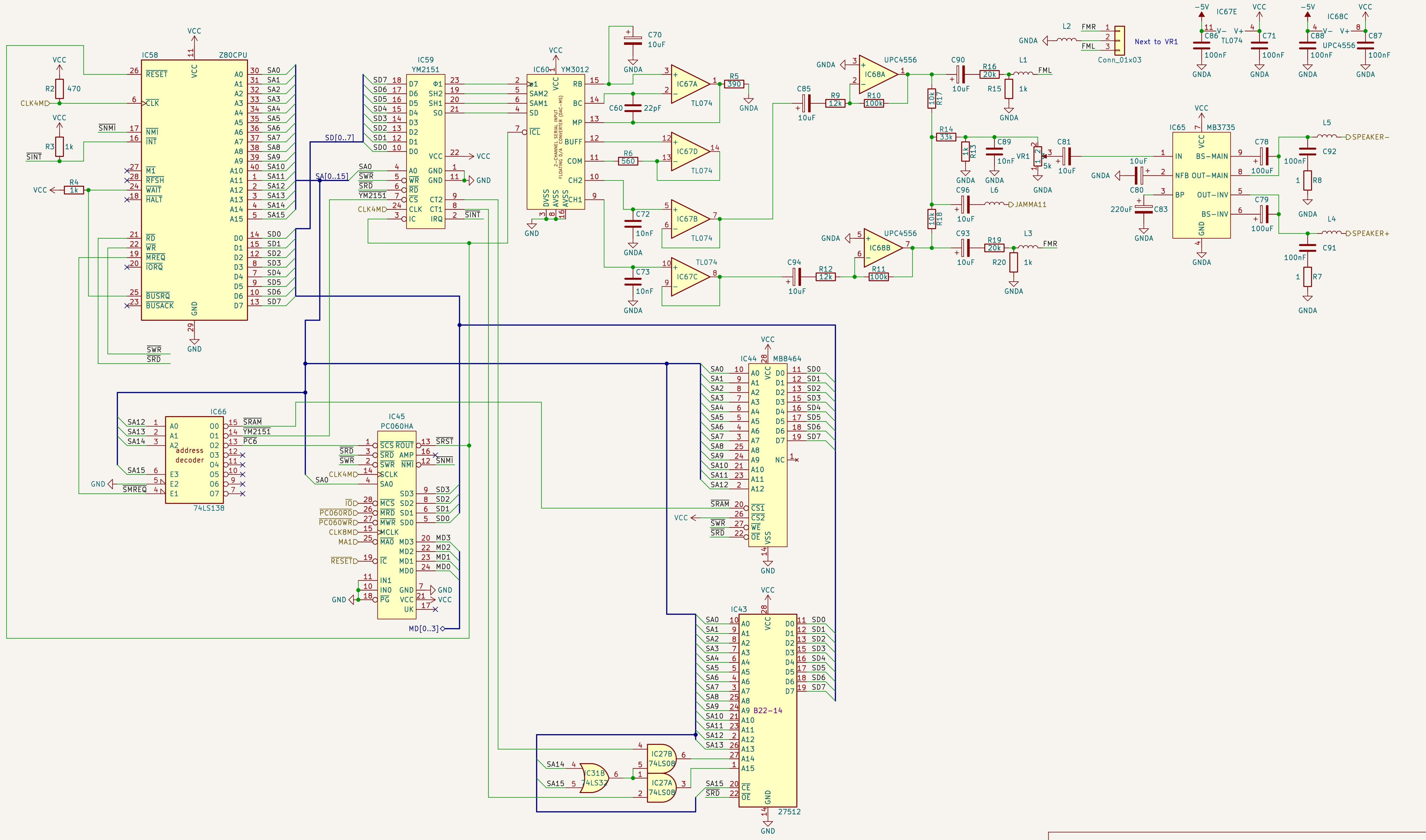
Taito K1100319A
Sheet: /IO/
File: io.kicad_sch

Title: Rainbow Islands Extra

Size: A3 Date: 2023-06-13 Rev:
KiCad E.D.A. 8.0.9 Id: 5/8



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Taito K1100319A

Sheet: /Sound/

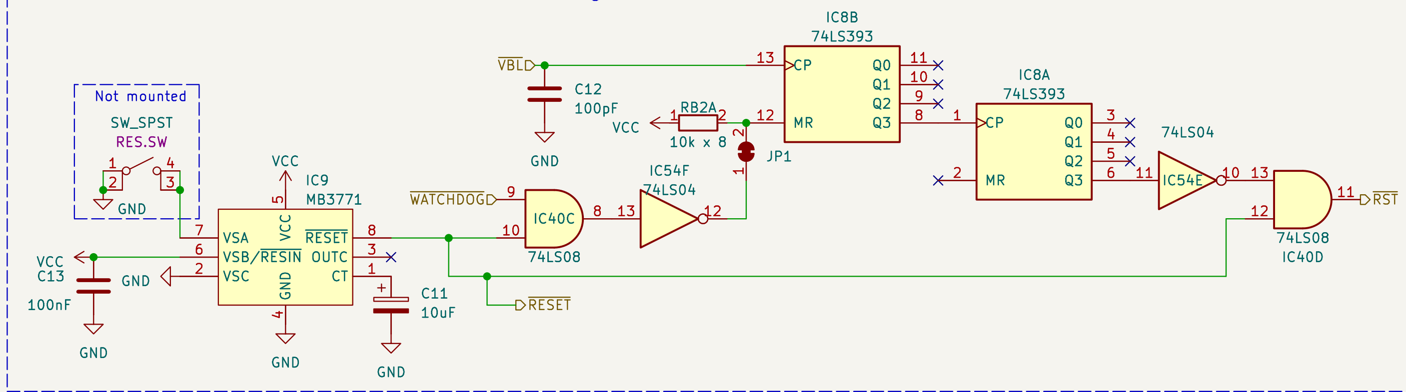
File: sound.kicad_sch

Title: Rainbow Islands Extra

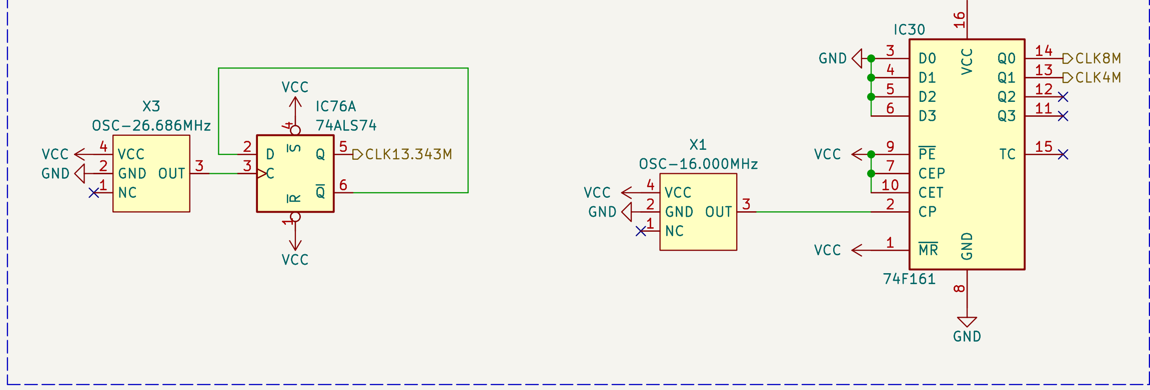
Size: A3 Date: 2023-06-13 Rev:

KiCad E.D.A. 8.0.9 Id: 6/8

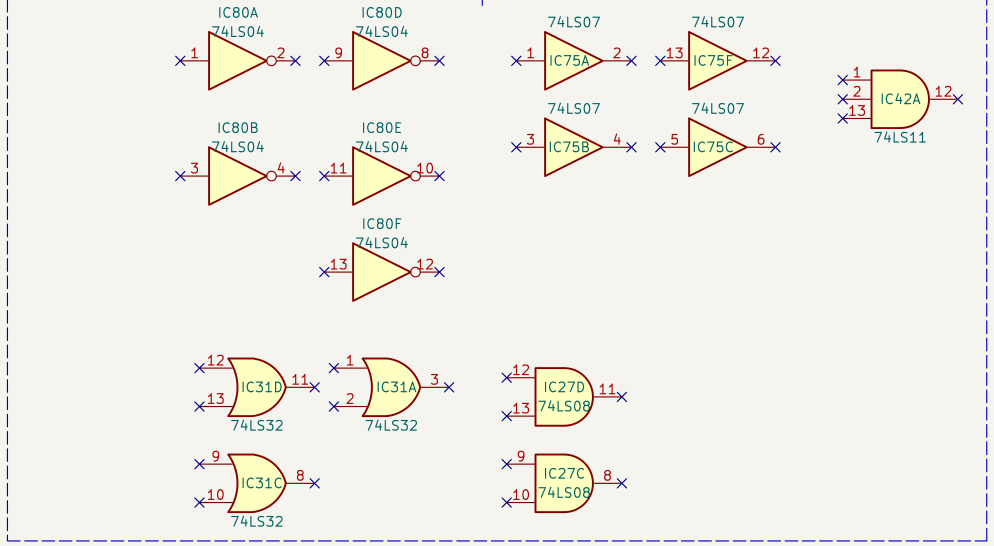
Watchdog and reset



Clocks



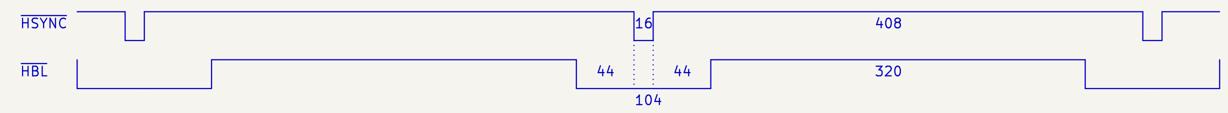
Spare



Horizontal and vertical synch timing diagrams

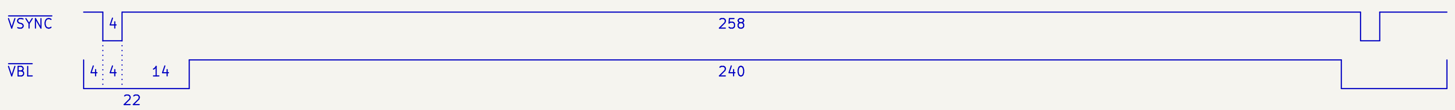
The pixel clock is derived from the 26.686MHz oscillator.
Pixel clock OVCK: f = 26.686MHz / 4 = 6.6715MHz

The numbers in the HSYNC and HBL diagram are pixel clock cycles.
All edges are synchronised to the rising edge of OVCK.



HSYNC and HBL
Frequency f = 6.6715MHz / 424 = 15.734670kHz.
Period T = 1/f = 63.5539us.

The numbers in the VSYNC and VBL diagram are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



VSYNC and VBL:
Frequency f = 15.734670kHz / 262 = 60.05599Hz
Period T = 1/f = 1 / 15.734670kHz = 16.6511ms

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Taito K1100319A
Sheet: /Misc/
File: misc.kicad_sch

Title: Rainbow Islands Extra

Size: A3 Date: 2023-06-13 Rev:
KiCad E.D.A. 8.0.9 Id: 7/8